

22425840

A)

```
?CRABS
```

Results

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b)

```
library(MASS)
data("crabs")
str(crabs)
```

c)

```
'data.frame':   200 obs. of  8 variables:
 $ sp      : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex     : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index   : int   1 2 3 4 5 6 7 8 9 10 ...
 $ FL      : num   8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW      : num   6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL      : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW      : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD      : num   7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

```
Freq_Tab_Species<-table(crabs$sp) ##frequency table species
Freq_Tab_sex<-table(crabs$sex) ##frequency table sex
```

Results

```
B O
F 50 50
M 50 50
```

The data is balanced across all the 4 groups. They have similar heights and width

Question 2

a)

```
?boxplot
boxplot(crabs$BD ~ crabs$sp+sex, data = crabs,
        main = "BD Defined by Sex and Species",
        xlab = "Species",
        ylab = "Body Depth")
```

```
outliers <- boxplot(crabs$BD~sp+sex, data = crabs, plot = FALSE)$out ###identifying outliers
outliers
```

Results

numeric(0)

Th2b) There are no outliers, there seem to be virtually no variability in the data

3)

```
?hist
par(mfrow = c(1, 2))
hist(crabs$CL[crabs$sp == "B"],
     main = " for Species B",
     xlab = "Carapace Length")
hist(crabs$CL[crabs$sp == "O"],
     main = "CL for Species O",
     xlab = "Carapace Length")
```

The distributions of the CL for the distributions are both normal (Hence the bell shaped, hence they are both mesokurtic. The distribution peaks around 30mm for B and between 35 to 40 of O.

4)

```
morph_mean <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
                        data = crabs, FUN = mean)

## grouped barplot of these means
par(mfrow = c(1, 1))
barplot(t(as.matrix(morph_mean[, -c(1,2)])),
       beside = TRUE, legend = TRUE,
       main = "Mean Morphological Measurements by Species and Sex",
       xlab = "Species and Sex",
       ylab = "Mean Value",
       names.arg = paste(morph_mean$sp, morph_mean$sex, sep = "-"))

## The greatest species difference and the least sex variation
species_diff <- apply(morph_mean[, -c(1,2)], 2, function(x) max(x) - min(x))
sex_var <- apply(morph_mean[, -c(1,2)], 2, function(x) abs(x[1] - x[2]))
species_diff
```

```

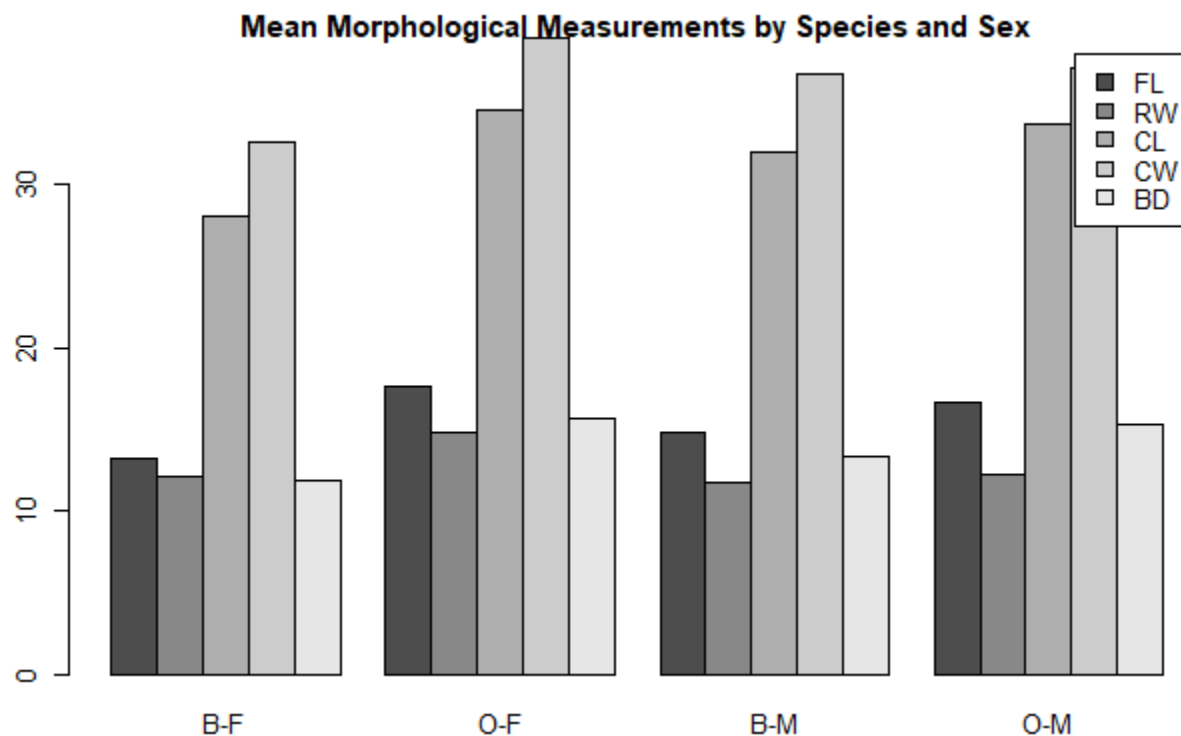
sex_var
variation <- apply(morph_mean[,-c(1,2)], 2,
                  function(x) abs(x[1] - x[2])/species_diff)
Compute the mean of each morphological
# measurement (FL, RW, CL, CW, BD) by species and sex
morph_mean <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
                        data = crabs, FUN = mean)
morph_mean

## The greatest species difference and the least sex variation
species_diff <- apply(morph_mean[,-c(1,2)], 2, function(x) max(x) - min(x))
sex_var <- apply(morph_mean[,-c(1,2)], 2, function(x) abs(x[1] - x[2]))
species_diff
sex_var
variation <- apply(morph_mean[,-c(1,2)], 2,
                  function(x) abs(x[1] - x[2])/species_diff)
variation

```

Results

sp	sex	FL	RW	CL	CW	BD
1	B F	13.270	12.138	28.102	32.624	11.816
2	O F	17.594	14.836	34.618	39.036	15.632
3	B M	14.842	11.718	32.014	36.810	13.350
4	O M	16.626	12.262	33.688	37.188	15.324



Question 5

```
pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],  
      main = "Scatterplot Matrix of Morphological Measurements",  
      pch = 19,)
```

Results

